

Australian Math Olympiad 2018 — P5/8

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The sequence $a_1, a_2, a_3, \dots$ is defined by $a_1 = 1$, and for $n \geq 2$,

$$a_n = (a_1 + a_2 + \dots + a_{n-1}) \times n.$$

Prove that a_{2018} is divisible by 2018^2 .

Solution

Let $n \geq 3$. Since $a_{n-1} = (a_1 + a_2 + \dots + a_{n-2})(n-1)$ we deduce that

$$a_n = a_{n-1} \frac{n}{n-1} + na_{n-1} = a_{n-1} \frac{n^2}{n-1}$$

but since a_{n-1} has a factor of $n-1$, we must have $n^2 \mid a_n$, and thus $2018^2 \mid a_{2018}$ as desired.